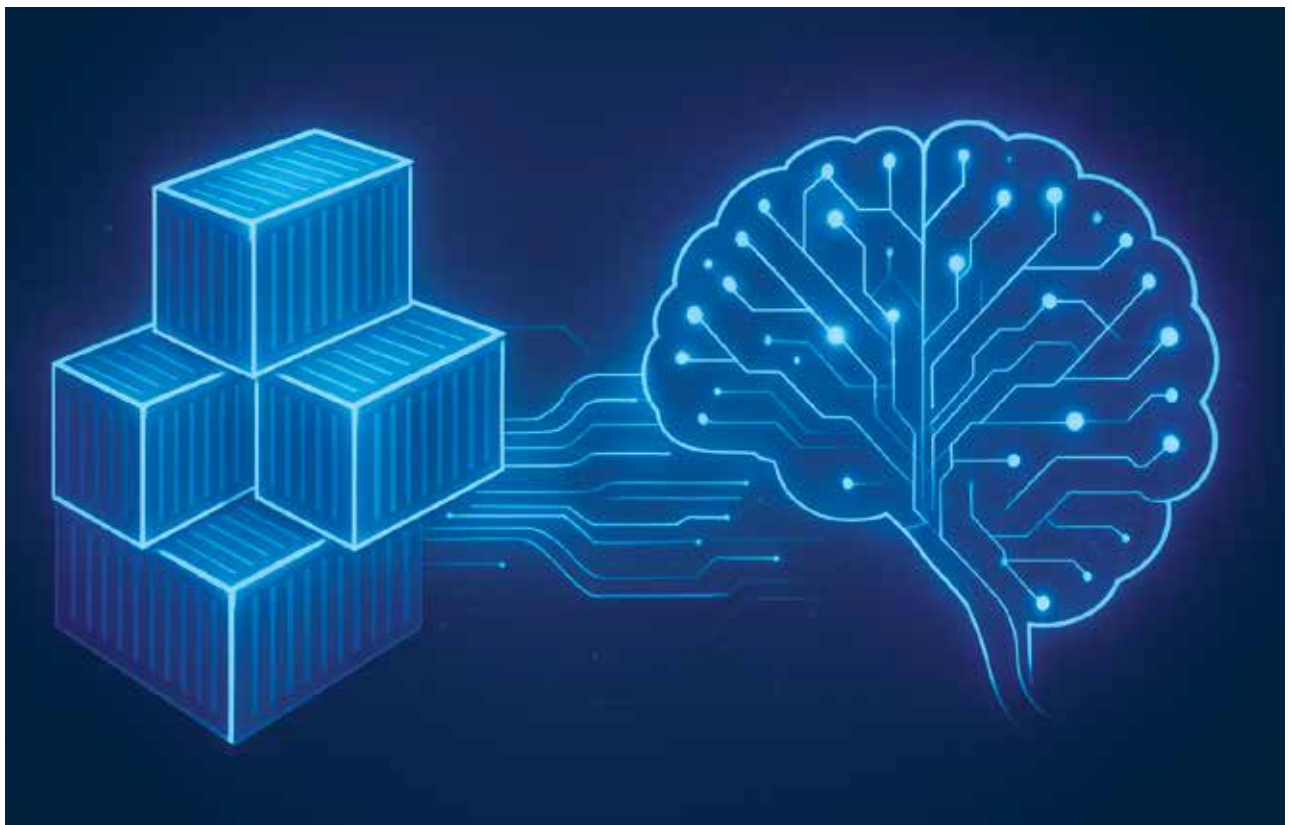


Docker Containers and Generative AI: A Powerful Partnership

The integration of Docker with generative AI makes it easier to create and deploy AI applications, especially those that use vector databases and large language models (LLMs).



Generative AI is automating content creation, code generation, image synthesis and other tasks across a variety of industries thanks to models like GPT and Stable Diffusion, as well as other large-scale neural networks. GenAI can also generate hypothetical scenarios along with real use cases using applications like Leonardo AI and Microsoft Designer (Figure 1).

Generative AI models often require:

- Key dependencies (CUDA, PyTorch, TensorFlow and Hugging Face libraries)

- Specific Python or operating system packages
- GPUs for efficient processing
- Consistent environments for training and inference

Integration of generative AI with Docker

A key factor in improving the scalability and deployment of generative AI systems is Docker, a potent containerisation platform. Docker guarantees consistent performance across a range of systems, from local computers to cloud platforms, by encapsulating AI models, dependencies and runtime